

- For ground-mounted sound barriers: The height of the top of the sound barrier above the lower surrounding ground surface.
- For structure- or traffic-barrier-mounted sound barriers: The height of the top of the sound barrier above the low ground or water surface surrounding the support structure.

In no case shall the structure height, Z , used in calculating K_Z be taken less than 33.0 ft.

Where the sound barrier is constructed directly atop an embankment, the height of the sound barrier should be measured from the lower ground surface surrounding the embankment.

Table 3.8.1.2.1-1—Gust Effect Factor, G

Structure Type	Gust Effect Factor, G
Sound Barriers	0.85
All other structures	1.00

Table 3.8.1.2.1-2—Drag Coefficient, C_D

		Drag Coefficient, C_D	
Component		Windward	Leeward
I-Girder and Box-Girder Bridge Superstructures		1.3	N/A
Trusses, Columns, and Arches	Sharp-Edged Member	2.0	1.0
	Round Member	1.0	0.5
Bridge Substructure		1.6	N/A
Sound Barriers		1.2	N/A

The term “columns” in Table 3.8.1.2.1-2 refers to columns in superstructures, such as spandrel columns in arches.

3.8.1.2.2—Loads on the Superstructure

In the general case of wind analysis, the wind load shall be determined as specified in Article 3.8.1.1 and the wind direction shall be varied. The wind loads shall be taken as the algebraic transverse and longitudinal components of the wind load. The wind direction for design shall be that which produces the maximum force effect in the component under investigation. The transverse and longitudinal components of the wind load shall be applied simultaneously.

C3.8.1.2.2

For superstructure components, the wind load on different members should be calculated separately and used in designing the members themselves. For trusses, the wind loads from different members and from the flooring system are transferred to the top and bottom planes of wind bracing and are used in designing the wind bracing system, including the end portals and cross-frames.

3.8.1.2.3—Loads on the Substructure

3.8.1.2.3a—Loads from the Superstructure

The transverse and longitudinal wind load components transmitted by the superstructure to the substructure for various angles of wind directions may be taken as the product of the skew coefficients specified in Table 3.8.1.2.3a-1, the wind pressure calculated using Eq. 3.8.1.2.1-1, and the depth of the bridge. The depth of the bridge shall be as seen in elevation perpendicular to the longitudinal axis of the bridge.

C3.8.1.2.3a

The Seventh Edition of *AASHTO Standard Specifications for Highway Bridges* was the first edition to incorporate the wind loads per skew angle of wind (Vincent, 1953). Alternatively, the wind load on the exposed area may be determined using an algebraic summation of transverse and longitudinal components of wind load. However, a wind directionality factor and drag coefficient other than the one specified in Article 3.8.1.2.1 should be considered.